GAME _ THEORY

* study of how players strategize & make decisions
* theoritical framework for conceiving social scenarios.

* Pure strategy :-
* Mixed strategy
* Depending on the no-of players
  - 2 players
  - N players.

* Zero sum game
  ↳ sum of gain & losses is zero.

A & B with m & n Strategies

$$
\begin{array}{c|c}
 & \begin{array}{cccc} B_1 & B_2 & \cdots & B_n \end{array} \\
\begin{array}{c} A_1 \\ A_2 \\ \vdots \\ A_m \end{array} &
\begin{bmatrix}
a_{11} & a_{12} & \cdots & a_{1n} \\
a_{21} & a_{22} & \cdots & a_{2n} \\
\vdots & \vdots & \ddots & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mn}
\end{bmatrix}
\end{array}
$$

Maximin value (maximize minimum gain) → $\underline{V}$.
Minimax value (minimize maximum gain). → $\overline{V}$

value of game is given by V: $\boxed{V = \underline{V} = \overline{V}}$

saddle point

A Game is fair when maximin = minimax value = 0

else strictly determinable.

~~obtain~~

P.T.O.

1. obtain payoff matrix
2. max (min{rows}) $\Rightarrow \underline{V}$
3. min (max{cols}) $= \overline{V}$
4. pure strategy is used, saddle point $V = \underline{V} = \overline{V}$
5. optimum strategy corresponds to the entry (A_i, B_j) for which value is V

Qn) 2 companies A & B sell 2 brands of flu medicine.

Company A advertises in Ratio

Radio (A1)
television (A2)
Newspaper (A3)

Company B, in addition to radio, using

Radio (B1)
Television (B2)
Newspaper (B3)
mails brochures (B4)

Depending on the effectiveness of each advertising campaign, one company can capture ~~to~~ a portion of the market from the other. The following matrix summarizes the % of the market captured or lost by company A.

Matrix:-

Company A.

	B1	B2	B3	B4
A1	8	-2	9	-3
A2	6	5	6	8
A3	-2	4	-9	5

$\underline{V} = \max(\{-3, 5, -9\}) \Rightarrow \underline{\underline{5}}$

$\overline{V} = \min(\{8, 5, 9, 18\}) = \underline{\underline{5}}$

$$\boxed{\underline{\underline{V = 5}} = \overline{V} = \underline{V}}$$

saddle point $= 5 + \rightarrow$ Gain for A loss for B.

~~Since the payoff matrix~~

Since the payoff matrix is for A & saddle point = 5, A gains 5 % of the market captured.

- The optimal soln of the gain occurs by selecting strategy 2 by A (A_2) & 2 by B (B_2) which means that both company should use television advertising.

Qn) Following is the payoff matrix in dollars for a 2 person zero sum game, using minimax criteria. Find the best strategy

	B1	B2	B3
A1	-3	5	-3
A2	1	3	5
A3	-3	-7	11

$\underline{V} = \max(\min\{\text{rows}\}) = \max\{-3, 1, -7\} = 1$

$\overline{V} = \min(\max\{\text{cols}\}) = \min\{1, 5, 11\} = 1$

$\underline{\underline{V = 1}}$

saddle point $= \underline{\underline{1}}$

strategy to be used A_2B_1 //.

Given

	B_1	B_2	B_3	max min
A_1	1	3	1	1
A_2	0	-4	-3	-4
A_3	1	5	-1	-1
max	1	5	1	$\boxed{1}$

4).

	B_1	B_2	max min
A_1	-1	6	-1
A_2	2	4	2
A_3	-2	-6	-6
min max	2	6	2

Player A pays ₹1 to B
Player B pays ₹6 to A
" B " ₹2 to A
B " ₹4 to A
A " ₹2 to B
A " ₹6 to A

strategy to be used $A_2 B_1$

{gain to A & loss to B}.

P.T.O

5.) Find range of values of P & Q which will render the entry (2,2) a saddle point for the game.

		Player B			max min
player 1	2	4	5		2
	10	7	Q		7
	4	P	6		4
min max	10	7	6	Imagine 7.	

soln: $Q \geq 7$ and $P \leq 7$

MIXED STRATEGY GAME:-

Graphical method → accepted only when $n \times 2$ and $2 \times n$ when matrix is $2 \times n$ or $n \times 2$.

1. A's expected payoff corresponding to B's jth pure strategy is :-

$$(a_{ij} - a_{2j})x_1 + a_{2j} \quad , j = 1, 2 \ldots n$$

2. Player A seeks the value of x_1 that maximizes the minimum expected ~~playoff~~ payoffs. That is :-

$$\boxed{\max_{x_i} \left\{ \min \left\{ (a_{1j} - a_{2j}) x_1 + a_{2j} \right\} \right\}}$$

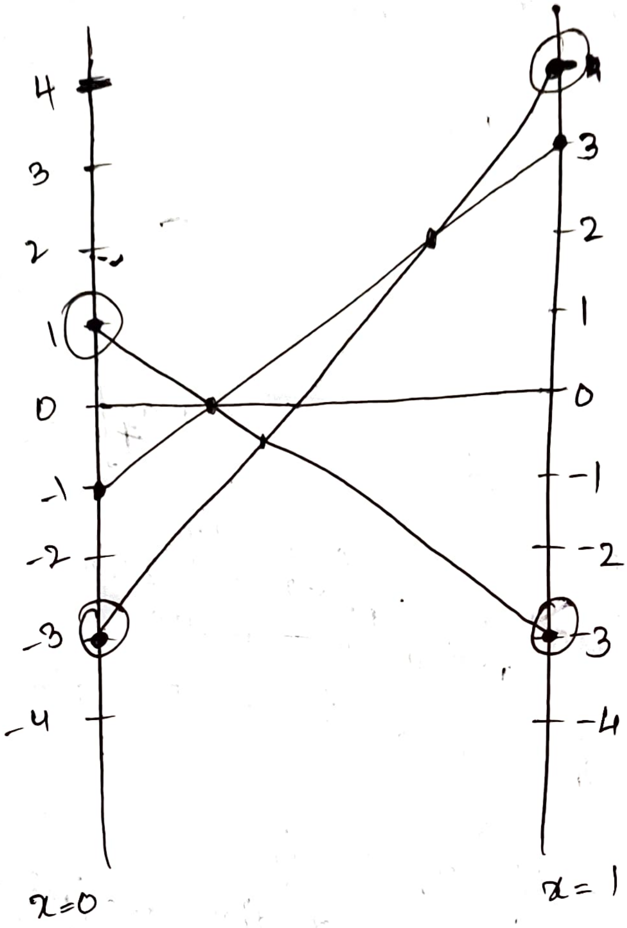
Q1) :- use graphical method to reduce the following game and hence solve :-

		y_1 B1	y_2 B2	y_3 B3	max min
					-3
(x_1)	A1	3	-3	4	-3
$(1-x_1)$	A2	-1	1	-3	
min max		3	1	4	

Construct graph:-

① draw $x = 0$ and $x = 1$ lines and mark all the labels/numbers.

② plot all points $A \longrightarrow x = 1$ line and $B \longrightarrow x = 0$ line and draw the line.



$$\begin{array}{ccc} x_1 & A_1 & \\ 1-x_1 & A_2 & \end{array} \begin{bmatrix} -3 & 4 \\ 1 & -3 \end{bmatrix}$$

$-3x_1 + 1(1-x_1)$

$\Rightarrow -3x_1 + 1 - x_1 \Rightarrow \boxed{1-4x_1}$

$4x_1 - 3(1-x_1)$

$4x_1 - 3 + 3x_1$

$\boxed{7x_1 - 3}$

Got 2 equations :- ~~equalize the~~

$$1 - 4x_1 = 7x_1 - 3$$

$$4 = 11x_1$$

$$\boxed{4/11 = x_1}$$

thus $1 - x_1 \Rightarrow 1 - \frac{4}{11} = \frac{7}{11}$

Therefore A uses strategy with probability 4/11 & strategy A2 with probability 7/11.

when $x_1 = 4/11$

$$1 - 4x_1 \Rightarrow 1 - \frac{16}{11} \Rightarrow \frac{11-16}{11} = -4/11$$

$$7x_1 - 3 \Rightarrow 7 \times \frac{4}{11} - 3 \Rightarrow \frac{28-33}{11} = \frac{-5}{11}$$

when $x = 7/11$.

$$1 - 4x_1 = 1 - \frac{28}{11} = \frac{11-28}{11} = \frac{-17}{11}$$

$$7x_1 - 3 \Rightarrow 7 \times \frac{7}{11} - 3 = \frac{49-33}{11} = \frac{16}{11}$$

$28 - 11 = 17$

$49 - 33 = 16$

soln :- y always looses

		y_1	y_2	y_3
		B_1	B_2	B_3
x_1	A_1	3	-3	4
$1-x_1$	A_2	-1	1	-3

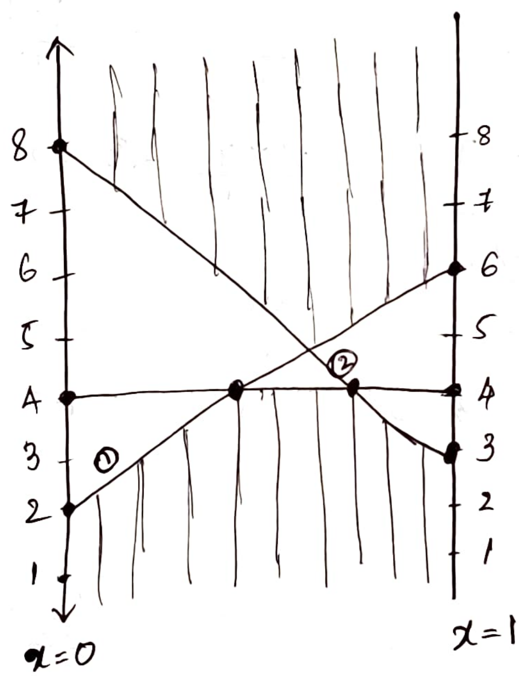
$\boxed{\text{Given } y_1 + y_2 + y_3 = 1}$

P.T.O.

Qn 2:-).

		B1	B2	B3	min
	A1	6	4	3	3
	A2	2	4	8	2
min	max	6	4	8	$\boxed{4}$

since minimax ≠ maximin, it is thus a mixed strategy.



Consider ① line corresponding to (6,2) and (4,4)

$$
\begin{array}{ccc}
 & & \begin{array}{cc} B_1 & B_2 \end{array} \\
x_1 & A_1 & \multirow{2}{*}{$\begin{bmatrix} 6 & 4 \\ 2 & 4 \end{bmatrix}$} \\
1-x_1 & A_2 &
\end{array}
$$

$$6x_1 + 2(1-x_1) = 4x_1 + 4(1-x_1)$$

$$6x_1 + 2 - 2x_1 = \cancel{4x_1} + 4 - \cancel{4x_1}$$

$$4x_1 = 2$$

$$\left\{ x_1 = 1/2 \right\}$$

$$x_2 = 1 - 1/2 = \underline{1/2}$$

Line (2) consider $\rightarrow$ (4,4) and (3,8)

$$\begin{array}{ccc} & & \\ x_1 & A_1 & \\ 1-x_1 & A_2 & \end{array}\begin{bmatrix} 4 & 3 \\ 4 & 8 \end{bmatrix}$$

$4x_1 + 2$
$4 \times \frac{1}{2} + 2 \Rightarrow \underline{\underline{4}}$

$\cancel{4x_1} + 4(1-\cancel{x_1}) = 3x_1 + 8 - 8x_1$

$4 = -5x_1 + 8.$

$-5x_1 + 8$
$-5 \times \frac{4}{5} + 8$
$\Rightarrow \underline{\underline{4}}$

$5x_1 = 8 - 4$

$5x_1 = 4$

$\underline{x_1 = 4/5}$ $\quad x_2 = 1/5$

Hence value of the game $= 4$, Because

$x_1 = 1/2 \quad 4x_1 + 2 \Rightarrow 4 \times \frac{1}{2} + 2 = \underline{\underline{4}}$

$x = 4/5 \quad -5x_1 + 8 \Rightarrow -5 \times \frac{4}{5} + 8 = \underline{\underline{4}}$

{Therefore A always gains 5 units.}

$y_1 \quad 1-y_1 \quad (y_3 = 0) \rightarrow$ (since y_3 is not present in the matrix)

$\begin{bmatrix} 6 & 4 \\ 2 & 4 \end{bmatrix} \rightarrow$ here $y_3 = 0$, thus $6y_1 + 4(\cancel{1}-y_1)$

$= 2y_1 + 4(1-y_1)$

$4y_1 = 0$

$\boxed{y_1 = 0}$

hence $y_2 = 1 - y_1$

$\boxed{y_2 = 1}$

optimal ~~strategy~~ for B $\underline{\underline{(0,1,0)}}$.

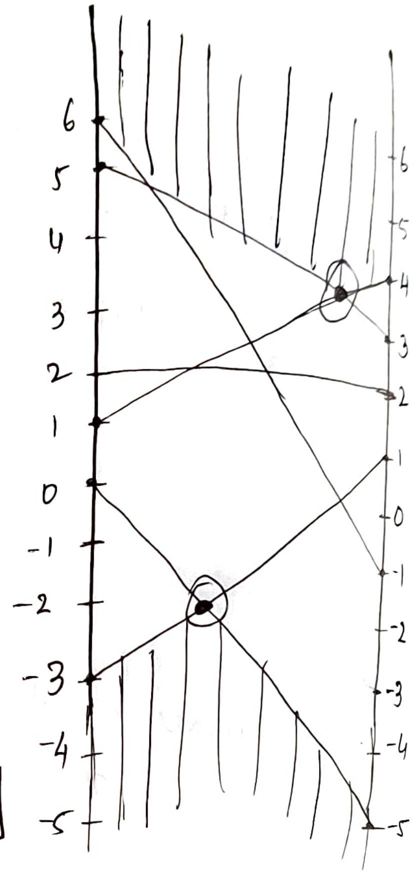
soln :- Optimal strategy for A $\Rightarrow$ (1/2, 1/2) or (4/5, 1/5)

Optimal strategy for B $\Rightarrow$ $\underline{\underline{(0,1,0)}}$

{Hence, B must be loosing 5 units}

Qn 3 :-). $n \times 2$ matrix

		B_1 (y_1)	B_2 $(1-y_1)$
(x_1)	A_1	1	-3
(x_2)	A_2	3	5
(x_3)	A_3	-1	6
(x_4)	A_4	4	1
(x_5)	A_5	2	2
(x_6)	A_6	-5	0



Soln :-

$$\boxed{x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 1}$$

$y_1 + y_2 = 1$

Hence $\boxed{y_2 = 1 - y_1}$

$2 \times n$:- lower envelop $\Rightarrow$ max
upper envelop $\Rightarrow$ min

$$\text{upper envelop :-} \begin{array}{c|cc} & y_1 & 1-y_1 \\ \hline x_2 & 3 & 5 \\ x_4 & 4 & 1 \end{array}$$

$$3y_1 + 5(1-y_1) = 4y_1 + 1(1-y_1)$$

$$3y_1 + 5 - 5y_1 = 4y_1 + 1 - y_1$$

$$5 - 2y_1 = 3y_1 + 1$$

$$4 = 5y_1$$

$$y_1 = 4/5$$

$$y_2 = 1/5$$

strategy for B $\Rightarrow$ $(4/5, 1/5)$

$$0 + 3x_2 + 0 + 4x_4 = 5x_2 + 1x_4$$

$$3x_2 + 4(1-x_2) = 5x_2 + 1 - x_2$$

$$3x_2 + 4 - 4x_2 = 4x_2 + 1.$$

$$4 - 1 = 8x_2 - 3x_2$$

$$\boxed{3/5 = x_2}$$

$$\boxed{x_4 = 2/5}$$

optimal strategy for A is :- $(0, 3/5, 0, 2/5, 0, 0)$

value of game $3x_2 + 4(1-x_2)$

$$3 \times \frac{3}{5} + 4\left(\frac{2}{5}\right)$$

$$\Rightarrow \frac{9+8}{5} = 17/5$$

$$3y_1 + 5(\overset{1-y_1}{y_2})$$

$$3 \times \frac{4}{5} + \frac{5}{5} = \frac{12+5}{5} = 17/5$$

$\boxed{\text{Thus the value is } 17/5}$

* **DOMINANCE METHOD**

used if not a pure strategy or mixed strategy of order $2 \times n$ or $n \times 2$.

P.T.O.

Qn:- Use dominance principle :-

$$\begin{array}{c|ccccc} & & & \text{player B} & & \\ \hline \text{player A} & 6 & 15 & 30 & 21 & 7 \\ & 3 & 3 & 6 & 6 & 4 \\ & 12 & 12 & 24 & 36 & 3 \end{array}$$

<u>check if</u> :- row i is less than i-1 by comparing each element. If it is less, eliminate the row.

Thus after elimination.

$$\begin{array}{ccccc} 1 & 2\times & 3\times & 4\times & 5 \\ 6 & 15 & 30 & 21 & 7 \\ 12 & 12 & 24 & 36 & 3 \end{array}$$

Eliminate col 2, column 3 and column 4, so it can be eliminated.

$$\begin{array}{c|cc} & y_1 & 1-y_1 \\ \hline x_1 & 6 & 7 \\ 1-x_1 & 12 & 3 \end{array}$$

$\Rightarrow$ After elimination row 2 and column 2, column 3, column 4.

$$6x_1 + 12 - 12x_1 = 7x_1 + 3 - 3x_1$$

$$12 - 6x_1 = 4x_1 + 3$$

$$9 = 10x_1$$

$$\boxed{x_1 = 9/10}$$

$$\boxed{x_3 = 1/10}$$

$$(9/10, 0, 1/10)$$

$y_5 = 1 - y_1.$

$$6y_1 + 7 - 7y_1 = 12y_1 + 3 - 3y_1$$
$$7 - y_1 = 9y_1 + 3$$
$$4 = 10y_1$$
$$4/10 = \underline{\underline{y_1}}$$

$9y_1 + 3$

$9 \times \frac{4}{10} + 3 \Rightarrow \frac{36+30}{10}$

$y_5 = \underline{\underline{6/10}}$

$7x_1 + 3 - 3x_1$

$4x_1 + 3 \Rightarrow 4 \times \frac{9}{10} + 3 \Rightarrow \frac{36+30}{10} = \frac{66}{10}$

$12 - 6x_1$

$12 - \frac{54}{10}$

$\Rightarrow \frac{120-54}{10} =$

$\boxed{\begin{array}{l} 120 \\ -54 \\ \hline 66 \end{array}}$

$\frac{70-4}{10} = \frac{66}{10}$

$\boxed{\text{Value of } A = 66/10}$, thus A gets a gain of 66/10 and B gets a loss of $-66/10$

P.T.O.

Qn :-).

$$\begin{array}{cc|cccc|c}
 & & & \text{player B} & y_3 & y_4 & \\
\hline
 & & 3 & 2 & 4 & 0 & x. \\
 & x_2 & 3^{x.} & \overset{x.}{4} & 2 & 4 & \\
\text{player A} & x_3 & 4 & 2 & 4 & 0 & \\
 & & 0 & 4 & 0 & 0 & x. \\
\hline
\end{array}$$

$$\begin{array}{c|cc|}
 & y_3 & 1-y_3 \\
\hline
x_2 & 2 & 4 \\
1-x_2 & 4 & 0 \\
\hline
\end{array}$$

$\Rightarrow \quad 2x_2 + 4 - 4x_2 = 4x_2$

$4 = 8x_2 - 2x_2$

$4 = 6x_2$

$\underline{\underline{x_2 = 4/6}}$

$x_3 = \underline{\underline{2/6}}$

$2y_3 + 4 - 4y_3 = 4y_3$

$4 = 8y_3 - 2y_3$

$\underline{\underline{4/6 = y_3}}$

$y_4 = \underline{\underline{2/6.}}$

$A = \underline{\underline{(0,\ 2/3,\ 1/3,\ 0)}}$

$\Rightarrow \quad 4x_2$

$B = \underline{\underline{\{0,\ 0,\ 2/3,\ 1/3\}}} \Rightarrow \quad 4 \times \frac{4}{6} = \underline{\underline{\cancel{16}^{8}/\cancel{6}_{3}}} = \underline{\underline{8/3}} \rightarrow$ value for A.

Qn :-

player B

$$\text{player A}\quad \begin{bmatrix} \cancel{3} & 2 & 4 & 0 \\ 3 & 4 & 2 & 4 \\ 4 & 2 & 4 & 0 \\ 0 & 4 & 0 & 8 \end{bmatrix}$$

X.

$$\begin{bmatrix} 4 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & \underbrace{0 \quad 8}_{avg} \end{bmatrix} \Rightarrow \begin{bmatrix} \cancel{4} & 3 \\ 2 & 2 \\ 4 & 4 \end{bmatrix} \text{ avg of } \begin{bmatrix} 2 & 4 \\ 4 & 0 \\ 0 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix} > \text{avg} \begin{array}{cc} \begin{array}{cc} y_3 & y_4 \end{array} & \\ \begin{bmatrix} 2 & 4 \\ 4 & 0 \\ 0 & 8 \end{bmatrix} & \begin{array}{c} x_2 \\ x_3 \\ x_4 . \end{array} \end{array}$$

consider

2 4 eliminate 1st row.

$\boxed{2 \quad 4}$

$$\begin{array}{cc} \begin{array}{cc} y_3 & y_4 \end{array} & \\ \begin{bmatrix} 4 & 0 \\ 0 & 8 \end{bmatrix} & \begin{array}{c} x_3 \\ x_4 \end{array} \end{array}$$

$$4x_3 + 0 = 0 + 8(1 - x_3)$$

$$= 8 - 8x_3$$

$4x_3$

$$12x_3 = 8$$

$$x_3 = \cancel{8}^{2} / \cancel{12}_{3}$$

$$x_3 = \underline{2/3} \qquad x_4 = 1/3$$

$$4y_3 = 8(1-y_3)$$
$$4y_3 = 8 - 8y_3$$
$$12y_3 = 8$$
$$y_3 = 8/12 = \underline{\underline{2/3}}$$

$$y_4 = \underline{\underline{1/3}}$$

$$4 \times 2/3 = \underline{\underline{8/3}}$$

$$A \longrightarrow \{0, 0, 2/3, 1/3\}$$
$$B \longrightarrow \{0, 0, 2/3, 1/3\}.$$

$$8(1-x_3)$$
$$8(1-2/3)$$
$$\Rightarrow 8 \times 1/3 \Rightarrow \underline{\underline{8/3}}$$

value of $A \Rightarrow \underbrace{8/3}_{\text{gain for A}}$

loss for $B = \underline{\underline{-8/3}}$